Evaluate yourself : 4

Write the overall redox reaction which takes place in the galvanic cell,

$Pt(s)\left|Fe^{2+}(aq),Fe^{3+}(aq)\right\|MnO_4^-(aq),\ H^+(aq),Mn^{2+}(aq)\left|Pt(s)\right.$

9.4 Thermodynamics of cell reactions

We have just learnt that in a galvanic cell, the chemical energy is converted into electrical energy. The electrical energy produced by the cell is equal to the product of the total charge of electrons and the emf of the cell which drives these electrons between the electrodes.

If 'n' is the number of moles of electrons exchanged between the oxidising and reducing agent in the overall cell reaction, then the electrical energy produced by the cell is given as below.

Electrical energy = Charge of 'n' mole of electrons $\times$ E_{cell} .....(9.20)

Charge of 1 mole of electrons = one Faraday (1F)

$\therefore$ Charge of 'n' mole of electrons = nF

Equation (9.20) $\Rightarrow$ Electrical energy = nFE_{cell} .....(9.21)

Charge of one elctron = 1.602×10^{-19} C

$\therefore$ Charge one mole of elctron = $6.023 \times 10^{23} \times 1.602 \times 10^{-19}$ C

= 96488 C

i.e., 1F $\simeq$ 96500 C

This energy is used to do the electric work. Therefore the maximum work that can be obtained from a galvanic cell is

$$(W_{max})_{cell} = - nFE_{cell} \quad .....(9.22)$$

Here the (-) sign is introduced to indicate that the work is done by the system on the surroundings.

We know from the Second Law of thermodynamics that the maximum work done by the system is equal to the change in the Gibbs free energy of the system.

$$\text{i.e.,} \quad W_{max} = \Delta G \quad .....(9.23)$$

From (9.22) and (9.23),

$$\Delta G = - nFE_{cell} \quad .....(9.24)$$

For a spontaneous cell reaction, the ΔG should be negative. The above expression (9.24) indicates that E_{cell} should be positive to get a negative ΔG value.

When all the cell components are in their standard state, the equation (9.24) becomes

$$\Delta G^o = - nFE^o_{cell} \quad .....(9.25)$$

We know that the standard free energy change is related to the equilibrium constant as per the following expression.

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